

English-Hindi Dataset Processing and Translation with LLM

Project Overview

- This project includes dataset preprocessing, filtering, and translation using Large Language Models.

Task 1: Dataset Processing

- Loaded train and validation Excel datasets in Google Colab.
- Merged datasets into a single DataFrame.
- Performed word count analysis for English and Hindi.
- Filtered sentences between 3 and 60 words.
- Calculated word count differences and retained rows between -10 and +10.
- Computed character counts and character differences.
- Exported cleaned dataset into Excel format.

Task 2: Translation with LLM

- Selected 200 English sentences from the cleaned dataset.
- Used LLaMA 3 / Mistral model for English to Hindi translation.
- Generated Hindi translations using prompt-based inference.
- Evaluated outputs using BLEU, CHRF, and TER metrics.
- Saved translations and evaluation scores.

Technologies Used

- Python
- Google Colab
- Pandas
- Transformers
- Hugging Face
- SacreBLEU
- OpenPyXL

Outputs

- Final_Cleaned_English_Hindi_Dataset.xlsx
- LLM_Translations.xlsx